

MODULE 3.2.S — DRUG SUBSTANCE

Amlodipine Besylate — AMLOZA Tablets ANDA

3.2.S.1 General Information

3.2.S.1.1 Nomenclature

INN	Amlodipine (as besylate salt)
Chemical name	3-Ethyl-5-methyl (±)-2-[(2-aminoethoxy)methyl]-4-(2-chlorophenyl)-1,4-dihydro-6-methyl-3,5-pyridinedicarboxylate, monobenzenesulphonate
CAS Registry Number	111470-99-6 (besylate salt)
Company/laboratory code	Not applicable — compendial (USP) drug substance

3.2.S.1.2 Structure

Molecular formula: C₂₀H₂₅ClN₂O₅·C₆H₆O₃S. Molecular weight: 567.1 g/mol. Amlodipine besylate contains one chiral center; it is marketed and used clinically as the racemate, consistent with the RLD (Norvasc).

3.2.S.1.3 General Properties

Description	White to pale-yellow crystalline powder
Solubility	Sparingly soluble in water; slightly soluble in ethanol; practically insoluble in non-polar solvents
pKa	~8.6 (basic amine); ~1.0 (weakly acidic, dihydropyridine ring)
Melting point	Approximately 199–201°C (with decomposition)
BCS classification	Literature reports amlodipine as exhibiting high aqueous solubility and high intestinal permeability, consistent with a Biopharmaceutics Classification System (BCS) Class I profile; however, this ANDA is supported by in vivo bioequivalence data (see Module 5) rather than a BCS-based biowaiver
Hygroscopicity	Non-hygroscopic under normal storage conditions

3.2.S.2 Manufacture

3.2.S.2.1 Manufacturer(s)

Amlodipine besylate drug substance is procured from a qualified, USFDA-registered API manufacturer holding an active Drug Master File (DMF). [Commercial API manufacturer name, FEI number, and DMF number to be inserted upon sourcing.]

3.2.S.2.2 Description of Manufacturing Process and Process Controls

The synthetic route, reaction steps, in-process controls, and process parameters are proprietary to the DMF holder and are referenced by Letter of Authorization (LOA) rather than reproduced in this ANDA, consistent with standard generic industry practice.

3.2.S.2.3 Control of Materials

Starting materials, reagents, and solvents used in synthesis are controlled per the DMF holder's internal specifications, referenced but not detailed in this application.

3.2.S.2.4 Controls of Critical Steps and Intermediates

Critical process steps and associated in-process controls (e.g., reaction completion, intermediate purity) are documented within the referenced DMF.

3.2.S.2.5 Process Validation and/or Evaluation

Process validation data for the drug substance synthesis resides with the DMF holder and is not independently generated by the ANDA applicant.

3.2.S.2.6 Manufacturing Process Development

Not applicable to the ANDA applicant; process development history is maintained by the DMF holder.

3.2.S.3 Characterisation

3.2.S.3.1 Elucidation of Structure and Other Characteristics

Structural confirmation (IR, ¹H-NMR, ¹³C-NMR, mass spectrometry, elemental analysis, X-ray diffraction for polymorph confirmation) is available from the DMF and from published pharmacopoeial reference data. Polymorphic form control is maintained by the drug substance manufacturer as part of the controlled crystallization process, with polymorph identity confirmed via X-ray powder diffraction (XRPD) at drug substance release, per the DMF. The ANDA applicant relies on the DMF holder's polymorph control strategy rather than independently generated polymorph screening data.

3.2.S.3.2 Impurities

Impurity control basis	USP monograph limits, supplemented by ICH Q3A(R2) thresholds for any non-compendial process-related impurities
Reporting threshold	0.05% (per ICH Q3A for drug substances with max daily dose ≤2 g)
Identification threshold	0.10%
Qualification threshold	0.15%
Known specified impurities	As listed in the USP Amlodipine Besylate monograph (e.g., Amlodipine Impurity A, B, C per USP nomenclature) — controlled within compendial limits

3.2.S.4 Control of Drug Substance

3.2.S.4.1 Specification

Description	White to pale-yellow crystalline powder
Identification	IR conforms to USP Reference Standard; HPLC retention time matches standard
Assay	98.0–102.0% (anhydrous basis), by HPLC
Related substances	Individual unspecified impurity $\leq 0.10\%$; total impurities $\leq 0.5\%$ (per USP/ICH)
Water content	$\leq 0.5\%$ w/w by Karl Fischer titration
Residual solvents	Within ICH Q3C Class 2/3 limits as applicable to the synthetic route
Elemental impurities	Controlled in accordance with ICH Q3D (Guideline for Elemental Impurities) and USP General Chapters <232> (Elemental Impurities — Limits) and <233> (Elemental Impurities — Procedures); a risk assessment covering the synthesis route, container closure system, and manufacturing equipment would be conducted to establish the control strategy, rather than reliance on the obsolete compendial heavy metals test
Residue on ignition	$\leq 0.1\%$
Particle size distribution	Monitored as a development/process parameter to support consistent blend flow and content uniformity during direct compression; not currently established as a formal release specification, as no PSD-dissolution correlation has been demonstrated for this BCS Class I compound. An in-house limit would be incorporated as a release specification only if supported by development data justifying it as a critical quality attribute

3.2.S.4.2 Analytical Procedures

Compendial USP methods are used for identification, assay, and related substances (HPLC with UV detection). Karl Fischer titration is used for water content. Any non-compendial method (e.g., particle size by laser diffraction) would be developed and validated in-house.

3.2.S.4.3 Validation of Analytical Procedures

Compendial methods are considered validated per USP General Chapter <1225> when used as written; verification (system suitability, specificity check against the site's actual instrumentation) is performed prior to use. Any in-house methods would be fully validated per ICH Q2(R1) for accuracy, precision, specificity, linearity, range, and robustness.

3.2.S.4.4 Batch Analyses

Certificate of Analysis (CoA) data from representative DMF-supplied batches would be presented here in a real submission, demonstrating consistent compliance with the proposed specification across multiple batches. [Placeholder — batch data pending actual API sourcing.]

3.2.S.4.5 Justification of Specification

The proposed specification is based directly on the current USP monograph for Amlodipine Besylate, supplemented where necessary with ICH Q3A/Q6A-aligned limits for parameters not addressed compendially (e.g., specific process-related impurities unique to the selected synthetic route).

3.2.S.5 Reference Standards or Materials

USP Amlodipine Besylate Reference Standard is used for identification and assay calibration. In-house working standards are qualified against the USP standard prior to routine use, per standard pharmacopoeial practice.

3.2.S.6 Container Closure System

Double low-density polyethylene (LDPE) bags, heat-sealed, packed within a fiber drum or HDPE container, consistent with DMF packaging specifications, providing adequate protection from moisture and light during transport and storage.

3.2.S.7 Stability

3.2.S.7.1 Stability Summary and Conclusions

A stability program per ICH Q1A(R2) is proposed, comprising long-term (25°C/60% RH) and accelerated (40°C/75% RH) storage conditions. No experimental stability data are available or presented in this academic dossier; in a real submission, the retest period would be established based on data generated by the DMF holder under this or an equivalent protocol.

3.2.S.7.2 Postapproval Stability Protocol and Commitment

Ongoing stability monitoring of the drug substance would follow the DMF holder's approved stability protocol, with annual confirmatory batches per ICH Q1A(R2).

3.2.S.7.3 Stability Data

Actual stability data tables (assay, impurities, water content trended over time/condition) would be presented here, sourced from the DMF. Not fabricated for this academic exercise.

Portfolio Note: This section demonstrates the correct ICH M4Q(R1) structure for 3.2.S and correctly reflects that, for a generic (ANDA) applicant sourcing API from a third-party DMF holder, most drug-substance-level manufacturing/process data is referenced by DMF/LOA rather than generated independently — this is standard and expected generic industry practice, not a gap.